Darryl Gwilliam
Operational Performance & Transformation Leader

Product, Innovations, Projects & Operational Transformation Portfolio

Delivering measurable improvements in safety, performance, and operational efficiency across complex industry & safety-critical environments.

Profile

Director-level leader specialising in operational performance, safety-critical systems, and transformation, with a strong focus on driving growth, continuous improvement, and measurable performance outcomes.

Proven track record delivering large-scale improvements across industry and complex environments through digital innovation, predictive maintenance strategies, and high-performance operating models.

Combines human factors, compliance, and stakeholder engagement to deliver safe, efficient, and reliable operations in high-risk, high-demand environments.

Currently developing a Human Performance Engine - a data-driven framework designed to optimise workforce performance, fatigue, competency, and operational decision-making, translating real-world challenges into scalable digital solutions.

Methodology

I apply a structured, engagement-led approach to operational transformation, ensuring solutions are practical, scalable, and aligned with real-world needs.

Key Strategic & Operational Case Studies

Case 1: *Growth*
Strategic Growth & Data Transformation of Infrastructure Operations

Situation

Operating under a multi-year contract with Iarnród Éireann (Irish Rail), the business relied heavily on traditional on-track machine (OTM) operations as its primary revenue source.

Emerging industry trends created clear strategic risks:

- Increasing focus on environmental performance and emissions reduction
- Pressure to maximise train availability on the network
- Shift towards data-driven maintenance and decision-making

These factors indicated a future where traditional maintenance volumes could decline, placing the existing business model at risk.

Action

I proactively identified this risk and developed a strategy to reposition the business towards a data-driven, higher-value operating model.

Strategic Insight

Recognised that long-term value would shift from:

- Volume-based maintenance → Data-led, precision-based operations

Identified opportunity to:

- Enter the rail data and surveying market
- Align directly with client strategic priorities

Business Model Transformation

Developed a dual-value model:

- Higher-quality, lower-volume maintenance
- New revenue stream through data services

Repositioned machines as:

- Not just maintenance tools → Mobile data acquisition platforms

Stakeholder Engagement

Engaged senior stakeholders within Irish Rail

- Aligned strategy with key priorities:
    - Reduced emissions
    - Increased network capacity
    - Improved safety

Successfully positioned a shift towards:

- Fewer interventions
- Higher quality outcomes

Operational Implementation

Introduced trial programmes integrating:

- Surveying capability into operations
- Operator upskilling for data collection

Developed Design Transposition approach:

- Every tamping shift became a data-driven design shift

Transitioned machines into:

- Hybrid maintenance + data platforms

Impact

Contract value increased from €50M → €150M

- Established new business unit: Digital Rail Services
- Maintained operational output while significantly increasing value per shift

Delivered client benefits:

- Improved safety (reduced manual inspection requirements)
- Increased network capacity (fewer track possessions)
- Reduced environmental impact through transition to HVO fuel

Strategic Value

- Demonstrates ability to anticipate industry change and act ahead of it
- Delivered business model transformation, not just operational improvement
- Strong stakeholder influence at senior and strategic levels
- Positioned organisation for a data-driven future in rail infrastructure

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Case 2: *Innovation & Efficiency*

High-Performance Operating Model for Continuous Production

Situation

Irish Rail set a challenge to operate a ballast cleaning machine continuously for 37 hours.

The existing operating model required:

- 3-hour maintenance shutdowns every 8 hours
- Significant loss of productivity and inefficient use of track access

Key constraints included:

- Mechanical reliability over extended operation
- Operator fatigue and exposure to dust and weather
- Lack of integration between maintenance and operations

Action

I led a full redesign of the operating model, focusing on both human performance and machine optimisation

Human-Centred Redesign

Re-engineered operator environment:

- Introduced camera systems enabling remote operation
- Relocated operators into enclosed, positive-pressure cabins
- Eliminated exposure to dust and harsh weather

Improved working conditions:

- Enhanced safety, comfort, and focus
- Enabled sustained performance over extended durations

Dynamic Maintenance Model

Replaced static maintenance with a risk-based, dynamic system:

- On-the-move monitoring
  - Continuous checks (temperature, visual inspections)
- Short "pit stop" interventions
  - 5–10 minute stops for critical checks
- Planned maintenance intervals
  - 20-minute structured inspections where required

Workforce Integration

Introduced technician-operator model:

- Combined operational and mechanical competencies
- Enabled real-time intervention and improved mechanical sympathy

System Optimisation

Enhanced decision-making through:

- Ground-penetrating radar data
- Improved visibility of ballast conditions

Increased efficiency through:

- Strategic tool placement across the machine
- Reduced downtime during interventions

Impact

- +60% increase in output
- ~40% reduction in delays (improved reliability and availability)
- Successful delivery of continuous 37-hour operation target

Additional outcomes:

- Improved machine condition through better handling and maintenance
- Enhanced operator wellbeing and performance
- Elimination of inefficient downtime cycles

Strategic Value

Demonstrates ability to redesign complex operational systems under constraint

- Strong integration of:
  - Human performance
  - Engineering
  - Data-driven decision-making
- Delivered step-change improvement in productivity and reliability
- Established a scalable model for high-performance rail operations

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Case 3: *Digital Transformation*

Digital Fit-for-Duty & Control System

Situation

Operating a geographically dispersed fleet, regulatory requirements mandated robust assurance that all personnel were fit for duty prior to commencing work.

Unlike traditional rail operations, there was no centralised booking-on location, with machines deployed nationwide. This created significant challenges:

- Lack of consistent, real-time fit-for-duty verification
- Inefficient reliance on proposed mobile supervisory checks
- Increased operational risk due to distributed workforce
- Limited visibility of workforce status and compliance

The regulator's expectation of physical oversight through mobile controllers presented a solution that was resource-intensive, inconsistent, and not scalable.

Action

I led the design and implementation of a fully integrated digital control and assurance system, replacing manual oversight with a scalable, real-time operational model.

Control Room Development

Established a 24/7 central control room acting as the operational hub for:

- Booking-on and booking-off processes
- Incident management and response
- Communication with client and field teams

Introduced hybrid controllers:

- Dual-skilled personnel with operational and driver competency
- Capable of supporting real-time decision-making and resource deployment

Digital Safety & Performance Stations

Designed and deployed a network of biometric-enabled safety stations across both depots and machines.

Key capabilities included:

- Biometric identification:
- Automatic user recognition upon interaction
- Personalised system engagement

Live video booking-on:

- Real-time visual fit-for-duty assessment via control room
- Direct communication between field staff and controllers
- Mandatory safety compliance:
- Forced review and sign-off of all safety notifications
- Elimination of missed critical communications

Impact

- Eliminated requirement for mobile controllers, significantly reducing operational inefficiency
- Delivered a scalable, regulator-compliant fit-for-duty assurance model
- Achieved full visibility of workforce status across a distributed network
- Strengthened safety communication, ensuring 100% engagement with critical updates
- Enhanced workforce performance through improved access to information and structured processes
- Recognised with an industry award from Irish Rail

Strategic Value

Demonstrates ability to translate regulatory requirements into scalable digital systems

Strong integration of:

- Human performance
- Compliance
- Operational control
- Established a foundation for future development of the Human Performance Engine

Showcases leadership in modernising workforce management within safety-critical environments

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Case 4: *Governance*

Regulatory Transformation & Workforce Optimisation – *Rulebook Change*

Situation

Rail operations in Ireland required two fully qualified drivers per machine, as mandated by national railway rules.
However, operational demand was highly variable throughout the year, creating significant challenges:

- Inefficient workforce utilisation during lower demand periods
- Resource shortages during peak operational periods
- Increased exposure to fatigue, sickness, and operational risk
- Requirement to overstaff to maintain resilience, driving unnecessary cost

The existing model lacked flexibility and placed pressure on both operational delivery and workforce sustainability.

Action

I led a full review of the operating model, challenging the necessity of the existing regulatory requirement and identifying opportunities to introduce a more efficient, risk-based approach.

Role Redefinition and Risk Assessment

Conducted a detailed analysis of the second driver role:

- Identified that the role primarily functioned as:
- Monitoring and oversight
- Contingency support
- Operational assistance

Developed a new role:

- Driver's Assistant - a competency-based role designed to safely fulfil support functions without requiring full driver qualification
- Undertook full risk assessment to validate safety and operational viability

Competency & Assurance Framework

Designed and implemented a standalone competency model:

- Task-specific training aligned to operational requirements
- Psychometric assessment to ensure suitability
- Defined competency standards and ongoing assessment
- Ensured full alignment with safety and compliance expectations

Regulatory & Stakeholder Engagement

Engaged extensively with key stakeholders:

- Irish Rail
- Commission Road and Rail
- Insurance and risk bodies

Presented a structured, evidence-based case for change:

- Risk-based justification
- Operational efficiency benefits
- Maintained safety integrity
- Built consensus and secured support across all parties

Rulebook Transformation

Led formal request to amend national railway rulebook

Successfully achieved regulatory approval to:

- Replace requirement for two fully qualified drivers
- Introduce Driver + Driver's Assistant model

Marked the first instance of an external contractor successfully influencing rulebook change at this level

Impact

- Significantly increased workforce flexibility across operations
- Enabled more efficient resource planning aligned to demand variability
- Reduced reliance on overstaffing, lowering operational costs
- Improved resilience to sickness, fatigue, and operational disruption
- Maintained full compliance with safety and regulatory standards

Strategic Value

Demonstrates ability to challenge and reshape established regulatory frameworks

Strong stakeholder engagement and influence at senior and regulatory levels

Delivered alignment between:

- Safety
- Compliance
- Commercial performance

Showcases leadership in delivering systemic change beyond operational constraints

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Case 5: Project Management

Predictive Operations & Environmental Risk Management - *Sand Mitigation Strategy for Desert Rail Operations*

Situation

During early operations of a heavy-haul railway in Abu Dhabi, recurring sandstorms posed a significant threat to operational reliability and safety.

Sand accumulation on the track resulted in:

- Train delays and operational disruption
- Increased safety risk where infrastructure became partially buried
- Reactive maintenance interventions with high resource demand

Early approaches focused on removing sand after it reached the railway, which proved inefficient, costly, and operationally disruptive.

Action

As Co-Chair of the Sand Mitigation Committee, I led a strategic shift from reactive clearance to a predictive, system-based approach.

Problem Reframing & Environmental Analysis

Challenged the conventional approach of improving sand removal capability
Recognised that intervention after sand reached the railway was too late

Led research into:

- Sand movement patterns
- Wind behaviour and seasonal trends
- Installed local weather monitoring stations to capture real-time environmental data

Predictive Risk Modelling

Developed a forward-looking risk model combining:

- Local weather station data
- External forecasting data sources

Enabled:

- 5–10 day visibility of wind conditions
- Identification of high-risk areas before impact
- Introduced a proactive alert system to trigger early intervention

Engineering & Mitigation Design

Designed location-specific mitigation solutions based on:

- Sand type
- Wind direction
- Terrain conditions

Implemented tailored mitigation measures including:

- Fencing and barriers to slow, divert, and trap sand
- Controlled sand accumulation zones to improve removal efficiency

Operational Integration

Redesigned maintenance strategy to align with predictive insights:

- Pre-positioning of resources ahead of sand events
- Targeted interventions in high-risk areas
- Reduction of reactive, emergency-based work

Impact

- Near elimination of sand-related delays and operational disruption
- Significant reduction in reactive maintenance and associated costs
- Improved safety by reducing exposure to extreme environmental conditions
- Enhanced planning accuracy and resource deployment
- Established a fully preventative, intelligence-led maintenance model

Strategic Value

- Demonstrates ability to shift operations from reactive to predictive systems
- Strong integration of environmental data, engineering, and operational planning
- Showcases strategic thinking in high-risk, complex environments
- Delivered scalable, long-term improvement in reliability and performance

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Core Capabilities

- Operational Transformation
- Strategic Growth & Business Model Development
- Performance Optimisation in Safety-Critical Environments
- Predictive Maintenance & Data-Driven Decision Making
- Human Factors & Workforce Performance
- Regulatory Engagement & Compliance Leadership
- Digital Systems Integration & Innovation

Proven ability to deliver measurable improvements in performance, safety, and commercial outcomes across complex operational environments.